

Employee Database

Normalization Analysis (1NF – 3NF)

1. Introduction

Normalization is the process of structuring a relational database to reduce data redundancy and eliminate undesirable insertion, update, and deletion anomalies. This document walks through the Employee Database schema and verifies that it satisfies the first three normal forms (1NF, 2NF, and 3NF), which is the standard target for most transactional application databases.

The schema consists of ten tables: departments, positions, employees, salaries, attendance, leave_types, leave_requests, performance_reviews, projects, and employee_projects.

2. First Normal Form (1NF)

A table is in 1NF when: every column holds a single, atomic value; there are no repeating groups or arrays within a column; and every row is uniquely identifiable (via a primary key).

How the schema satisfies 1NF:

- Every table has a single-column (or composite) primary key that uniquely identifies each row — e.g. **employee_id** in employees, or the composite key (employee_id, project_id) in employee_projects.
- No column stores multiple values. For example, an employee's leave records are not stored as a comma-separated list inside the employees table — they live in their own leave_requests table, one row per request.
- Similarly, project assignments are not stored as a list inside employees; the employee_projects junction table gives each employee–project pairing its own row.
- Address fields, names, and statuses are stored as single atomic values rather than combined or encoded strings.

Conclusion: All ten tables satisfy 1NF.

3. Second Normal Form (2NF)

A table is in 2NF when it is already in 1NF and every non-key attribute is fully functionally dependent on the *entire* primary key — i.e. no non-key column depends on only part of a composite key. Tables with a single-column primary key automatically satisfy 2NF once they are in 1NF, so the main check applies to tables with composite keys.

Relevant case — employee_projects:

This table has a composite primary key of (employee_id, project_id). Its only non-key attributes are role_in_project and assigned_date, both of which describe the specific pairing of that employee with that project — not the employee alone, and not the project alone. There is no partial dependency.

Conclusion: All tables satisfy 2NF.

4. Third Normal Form (3NF)

A table is in 3NF when it is already in 2NF and contains no transitive dependencies — that is, no non-key column depends on another non-key column rather than directly on the primary key.

Design decisions made specifically to satisfy 3NF:

- **Departments extracted from employees.** department_name and location depend on department_id, not directly on employee_id. Storing them inside the employees table would mean department_name is transitively dependent on employee_id through department_id. Instead, employees stores only the department_id foreign key, and department details live in the departments table.
- **Positions extracted from employees.** Likewise, a job title's salary band (min_salary, max_salary) depends on position_id, not on any individual employee, so it is kept in its own positions table rather than repeated on every employee row.
- **Salary history separated from employees.** Rather than storing a single current salary directly on the employee row (which would need to be overwritten on every raise, destroying history), base_salary and effective_date are stored in a separate salaries table keyed by salary_id, with employee_id as a foreign key. This also avoids a transitive dependency between salary and other employee attributes.
- **Leave type details separated from leave requests.** type_name and max_days_per_year depend on leave_type_id, not on the specific leave request, so they are kept in leave_types rather than duplicated on every leave_requests row.
- **Project details separated from the assignment table.** project_name and status depend on project_id alone, so they live in projects; employee_projects stores only the foreign keys plus attributes that describe the assignment itself.

Conclusion: All tables satisfy 3NF — no non-key column is functionally dependent on another non-key column.

5. Table-by-Table Summary

Table	Primary key	1NF	2NF	3NF	Notes
departments	department_id	Yes	Yes	Yes	No transitive dependencies.
positions	position_id	Yes	Yes	Yes	department_id is a plain FK reference.
employees	employee_id	Yes	Yes	Yes	Department/position details kept out via FKs.
salaries	salary_id	Yes	Yes	Yes	Historical rows avoid overwrite anomalies.
attendance	attendance_id	Yes	Yes	Yes	One row per employee per work date.
leave_types	leave_type_id	Yes	Yes	Yes	Lookup table for leave categories.
leave_requests	leave_id	Yes	Yes	Yes	leave_type details not duplicated here.
performance_reviews	review_id	Yes	Yes	Yes	reviewer_id is a self-referencing FK.
projects	project_id	Yes	Yes	Yes	department_id is a plain FK reference.
employee_projects	(employee_id, project_id)	Yes	Yes	Yes	Composite key; junction table for the M:N relationship.

6. Anomalies This Design Avoids

- **Update anomaly:** Changing a department's name only requires updating one row in departments, not every employee row in that department.
- **Insertion anomaly:** A new department or position can be added without needing to also insert a dummy employee record.
- **Deletion anomaly:** Removing the last employee from a department does not delete the department's information, since department data lives independently in its own table.
- **Loss of history:** Because salaries are stored as a time-stamped history rather than a single mutable column, past compensation is preserved rather than overwritten.

7. Beyond 3NF

Boyce–Codd Normal Form (BCNF) and higher normal forms (4NF, 5NF) address edge cases involving overlapping candidate keys or multi-valued dependencies. None of the tables in this schema have multiple overlapping candidate keys or independent multi-valued facts about the same key, so 3NF is also equivalent to BCNF here. Third normal form is sufficient for this schema's purposes.

This document accompanies the Employee Database schema (SQL script, Word structure document, and ER diagram).